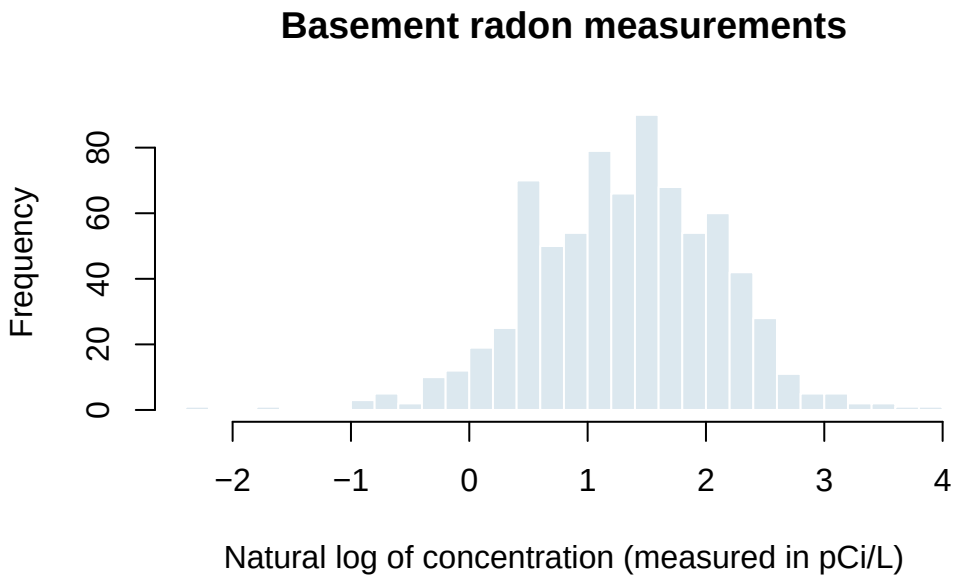
Lab 8: Partial Pooling and Gibbs Sampling

Sophie Xu

A. Read the data and model

1.

```
   county   n raw_average
   MURRAY   1      2.4932
 ST LOUIS 100      0.8461
```




```
pdf
  2
```

The histogram shows log radon for individual homes ranging from roughly -1 to 3.5.

In the model, within-county variation σ is the spread of individual homes around their own county's mean θ_j, while between-county variation τ is the spread of the county means θ_j around the shared center μ.

Murray's has one measurement from a single home, so it has sampling variance σ^2 around θ_j and could reflect one unusual house rather than the county's typical level. St. Louis's average of 0.846 is based on 100 homes, so its variance is only $\sigma^2/100$ and it is a much more reliable estimate of that county's mean.

B. Work through one update

```
   county data_weight conditional_mean
   MURRAY      0.2462           1.7445
 ST LOUIS      0.9703           0.8655
```

2.

County	n_j	$\bar{y}_j$	w_j	m_j
MURRAY	1	2.493	0.246	1.744
ST LOUIS	100	0.846	0.970	0.865

Murray: $w_j = \frac{1(0.16)}{0.49+1(0.16)} = 0.246$, $m_j = 0.246(2.493) + 0.754(1.5) = 1.744$.

St. Louis: $w_j = \frac{100(0.16)}{0.49+100(0.16)} = 0.970$, $m_j = 0.970(0.846) + 0.030(1.5) = 0.865$.

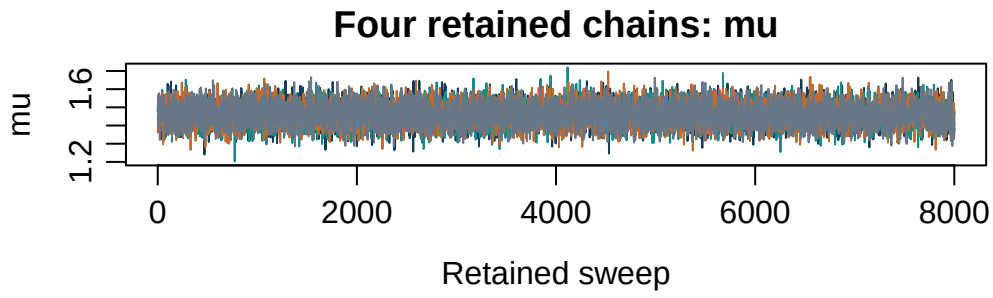
The St. Louis update gives more weight to the county data because its average is based on 100 homes.

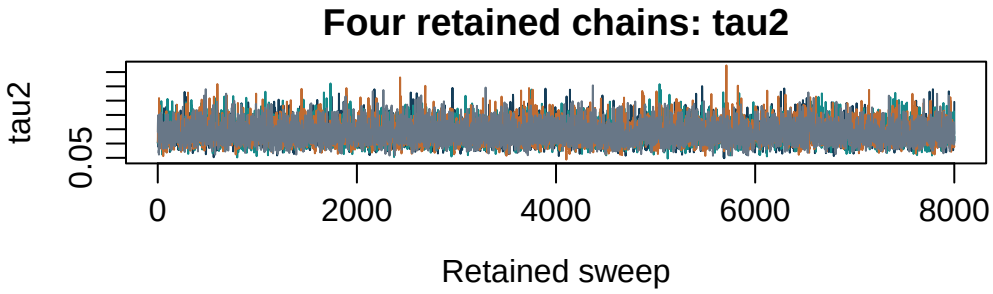
Murray's conditional mean is closer to the shared center, since with only one home its raw average is pulled most of the way back toward μ.

C. Run and understand the sampler

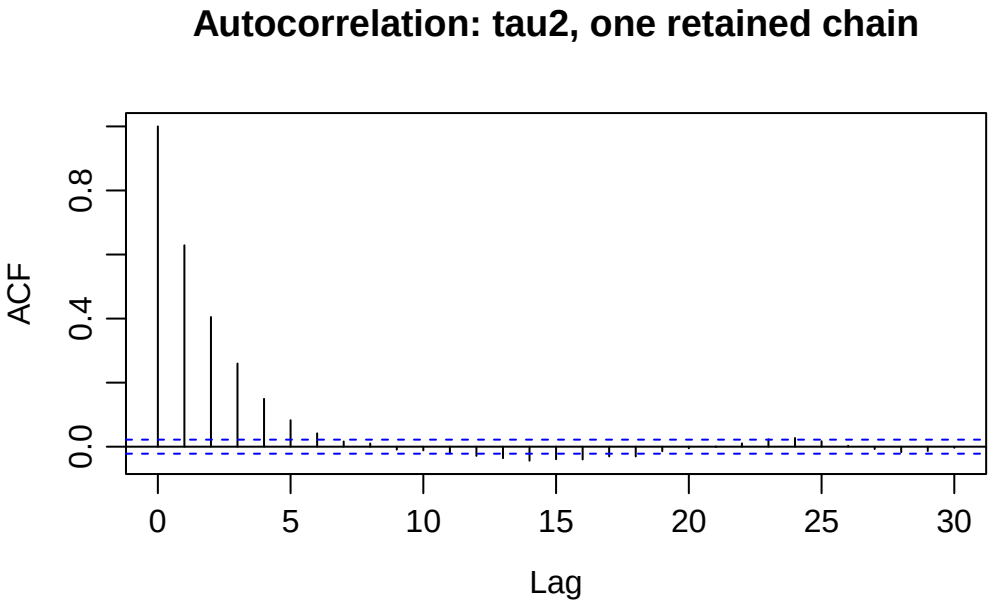
3.

We draw `theta` from its conditional distribution instead of setting it equal to `mtheta` because Gibbs sampling requires a random draw from each full conditional. Setting $\theta_j = m_j$ would ignore the conditional uncertainty V_j, so the saved draws would not represent the posterior and would understate uncertainty about the county means. We use `sqrt(Vtheta)` because `rnorm()` takes a standard deviation, while V_j is a variance.


Four retained chains: mu
mu
1.2
1.6
0
2000
4000
6000
8000
Retained sweep


Four retained chains: tau2
tau2
0.05
0
2000
4000
6000
8000
Retained sweep

```
pdf
  2
```


Autocorrelation: tau2, one retained chain
ACF
0.0
0.4
0.8
0
5
10
15
20
25
30
Lag

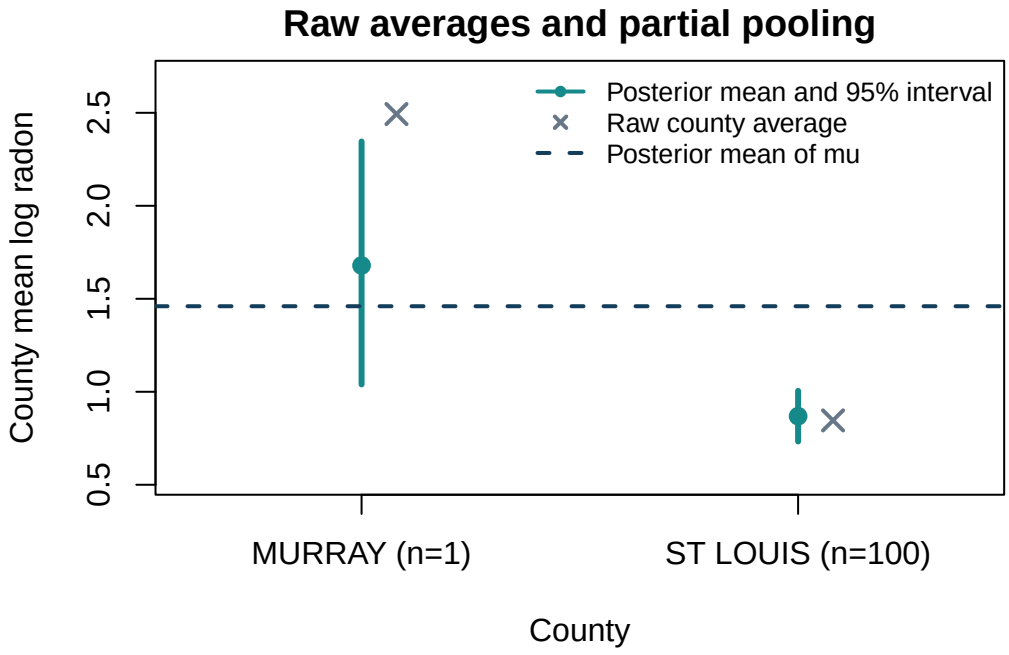
```
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```

4.

Yes, the four chains explore the same region. Although they started at very different values, after burn-in their traces overlap completely: μ moves within roughly 1.3 to 1.6, and τ^2 mostly stays between about 0.05 and 0.25. Therefore, the retained draws from all four chains appear to come from the same posterior distribution.

D. Compare the two counties

```
   county   n raw_average posterior_mean    low  high
   MURRAY   1      2.4932         1.6791 1.0395 2.346
ST LOUIS 100      0.8461         0.8689 0.7328 1.005
```




```
pdf
  2
```

County	Raw average	Posterior mean	95% interval
MURRAY	2.493	1.679	[1.039, 2.35]
ST LOUIS	0.846	0.869	[0.733, 1.00]

5.

Murray's estimate changes much more from its raw average: it moves from 2.493 to 1.679, while St. Louis moves only from 0.846 to 0.869. This matches the weights from Part B: with $n_j = 1$, Murray's data weight was only $w_j = 0.246$, so its estimate is pulled most of the way toward μ. While St. Louis, with $n_j = 100$, had $w_j = 0.970$, so its estimate stays close to its own average.

Murray's interval is also much wider. The conditional variance shrinks as n_j grows: using the Part B values, $V_j = 0.1206$ for Murray but only 0.00475 for St. Louis, so one measurement leaves much more uncertainty about θ_j than 100 measurements do. The full sampler's results differ slightly from Part B (e.g., Murray's posterior mean is 1.679 rather than 1.744) because it averages over uncertainty in μ, σ^2, and τ^2 instead of holding them fixed.

E. Interpret the results

6.

Sharing information across counties lets Murray's estimate borrow strength from the other 84 counties: instead of relying on one home's reading of 2.493, the model shrinks it toward the shared center μ (about 1.46), giving a posterior mean of 1.679 with an interval of $[1.039, 2.35]$ that shows how little one measurement reveals.

A 95% interval for θ_j would not contain 95% of future home measurements, because it describes uncertainty about the county's mean log radon, while individual homes vary around θ_j with within-county variance σ^2, so an interval for a new home measurement would need to add that variation and be much wider.

Agreement among the chains shows only that the sampler converged to the posterior implied by our model; it does not establish that the Normal model describes the data well, which would require separately checking the model's fit, such as comparing the observed data to what the model predicts.